Electron Layer and Leapfrogging Unknots

A Registered Numerical Falsifier – v0.1.0

SST-Workbench scientific pilot

28 October 2026

DEVELOPMENT – DIAGNOSTIC ONLY. Reduced ring induction is a mathematical diagnostic, not a validated three-dimensional finite-core Euler solution, electron, spin-1/2 state, or discrete quantum layer. Framework v1.1.0 source embedding/certification must be performed with its original release generator.

1 Identification, scope and status

Research family: new, official catalog ID not yet assigned; no existing SST-Workbench family is overwritten. Project: `electron_layer_leapfrogging`, version `v0.1.0`. Parent design: `preregistration/ORIGINAL_DESIGN.md`. Framework integration target: v1.1.0, self-contained runtime generated locally from verified source. Current scientific implementation: one circular filament, two equal-sign filaments, two opposite-sign filaments and global circulation reversal. This report has been authored before any physical promotion.

2 Falsifiable question and objective

Can correlated coaxial unknotted ring dynamics reproduce convergent classical order exchanges without a quantum-layer interpretation; and does extending the system to independent toroidal/poloidal material circulations impose a spinorial $2\pi/4\pi$ response *without* separately postulating a spin representation? The present version computes only the first, classical control. The latter question requires an independent fully coupled Euler field and configuration-space topological analysis, currently absent.

3 Competing hypotheses

H_{0a}: rotation and circulation signs are classical pseudovector orientations. H_{0b}: geometric rank exchange is not an intrinsic discrete energy level. H_{0c}: real Euler field values are single-valued under 2π spatial rotation; any projective sign or exchange statistics requires an independently derived state space and representation. H_{1a}: after full-field convergence there are non-artificial robust discrete internal eigenclasses. H_{1b}: a physically selected state representation has holonomy -1 after 2π and $+1$ after 4π. H_1 cannot be supported by the present reduced solver.

4 Assumptions, validity domain and edge cases

Unforced inviscid incompressible Euler is the target theory, but the implemented reduced model is *not* its discretization: coaxial circular vortex lines are constrained to remain circular; cross-section evolution follows an imposed a^2R proxy; and self-induced speed is approximated by a slender-core formula. There is no actual finite-core vorticity transport, no independent toroidal flow degree of freedom, no pressure Poisson solve, no vortex stretching or reconnection and no externally prescribed vertical energy potential. Out-of-domain large a/R, tube contact, non-axisymmetric instabilities, or numerical collapse block interpretation rather than produce a physical negative claim.

5 Symbols, units and dimensional audit

Symbol	Meaning	SI unit	Dimensions
R_i	major radius	m	L
z_i	axial center coordinate	m	L
a_i	regularizing/core proxy radius	m	L
Γ_i	signed circulation	$\mathrm{m}^2\ \mathrm{s}^{-1}$	L^2T^{-1}
ρ_f	effective fluid density	$\mathrm{kg}\ \mathrm{m}^{-3}$	ML^{-3}
t	absolute evolution parameter	s	T
Q_z	transverse-section flux	$\mathrm{m}^3\ \mathrm{s}^{-1}$	L^3T^{-1}

The length unit is $R_0 = r_c = 1.40897017 \times 10^{-15}$ m; characteristic speed is $\|\mathbf{v}_\circlearrowleft\| = 1.09384563 \times 10^6$ m s^{-1}. *Only as conditional SST scales*, set $\Gamma_0 = 2\pi R_0 \|\mathbf{v}_\circlearrowleft\|$ and $t_0 = R_0^2/\Gamma_0$. All numerical integration uses dimensionless $R/R_0, z/R_0, t/t_0, \Gamma/\Gamma_0$. These are not independently predicted electron properties.

6 Exact mathematical equations and observables

E1: two-ring interaction. Let source ring j be $\mathbf{X}_j(\varphi) = (R_j \cos\varphi, R_j \sin\varphi, z_j)$, evaluated at $\mathbf{x}_i = (R_i, 0, z_i)$. For $i \neq j$,

$$\mathbf{u}_{i\leftarrow j} = \frac{\Gamma_j}{4\pi} \int_0^{2\pi} \frac{\mathbf{X}_j'(\varphi) \times (\mathbf{x}_i - \mathbf{X}_j(\varphi))}{\left(\|\mathbf{x}_i - \mathbf{X}_j(\varphi)\|^2 + a_j^2\right)^{3/2}}\, d\varphi. \tag{1}$$

This is a *specified softened kernel*, not the exact Euler Green's function of a resolved core. Velocity dimensions: $[\Gamma][\mathbf{X}'][\mathbf{r}]/[r]^3 = L/T$.

E2: self translation (closure).

$$U_i^{\mathrm{self}} = \frac{\Gamma_i}{4\pi R_i}\left[\log\left(\frac{8R_i}{a_i}\right) - \frac{1}{4}\right], \quad \dot{R}_i = \sum_{j\neq i} u_{r,i\leftarrow j}, \quad \dot{z}_i = U_i^{\mathrm{self}} + \sum_{j\neq i} u_{z,i\leftarrow j}. \tag{2}$$

The 1/4 coefficient is a conventional model choice, not a fitted SST electron constant. The coupling of a soft mutual kernel and a slender-core self formula is an approximate composite closure.

E3: material-volume proxy and E4: order events.

$$a_i^2(t)R_i(t) = a_i^2(0)R_i(0), \qquad \Delta z(t) = z_2(t) - z_1(t), \qquad N_{\mathrm{swap}} = \#\{t : \Delta z(t) = 0 \text{ with a sign change}\}. \tag{3}$$

$[a^2R] = L^3$, $[\Delta z] = L$ and N_{swap} is dimensionless. In particular N_{swap} is NOT an electron orbital principal quantum number.

Numerical scheme. Source angle midpoint/equispaced trapezoidal sum with n_φ points; classical fixed-step fourth-order Runge–Kutta for the reduced ordinary differential equations. No grid-based fluid conservation, finite-core Euler convergence or strain diagnostic can be inferred from these ODE tests.

7 Equation-to-code sequence

First load and hash `configs/pilot.json`; generate circular analytic inputs; calculate Python float64 E1–E3 using `science/ring_model.py`; compute E4 with `axial_order_crossings`; independently run the native C++17 kernel in `native/ring_core.hpp` through the CLI for local parity and through `science._ring_native` for actual Framework scientific backend after installation; perform fixed-t_{final} refinements of quadrature and time step separately; write raw traces, ledgers and immutable hashes. Only actual execution may produce a backend record.

8 Sources, provenance, independence

The primary pilot uses deterministic, explicitly enumerated analytic circular rings. No E010 atlas is copied or used and no discovery target labels are imported. Sign flips and mesh/time refinements are numerical controls, not independent experimental evidence. The 3D Euler field provider, external data and independently resolved ring internal twist do not yet exist in the current package; their scientific claims must remain unresolved. Geometries are metric only after conditional choice of R_0.

9 Gate map and decision semantics

Gate	Question	Present status ceiling
P01	Single-ring self-speed and imposed circulation/core proxy	DIAGNOSTIC
P02	Three or more robust overtakes, convergence, full-field validity	UNRESOLVED
P03	Opposite circulation and orientation reversal	DIAGNOSTIC
P04	Independent poloidal/toroidal material circulation dynamics	UNRESOLVED
P05	Internal spectral levels beyond order swaps	UNRESOLVED
P06	Spinorial $2\pi/4\pi$ and exchange holonomy	UNRESOLVED
P07	Volume-balanced jets and no electromagnetic projection	UNRESOLVED
G0	Actual Framework source integrity and freeze	PENDING
G1–G2	Analytic data and reduced-model diagnostics	DIAGNOSTIC
G3–G4	Blind discovery and independent full Euler confirmation	UNRESOLVED
G5–G6	Actual native pybind11 kernel; GPU only if executed	PENDING/UNRESOLVED
G7–G8	Independent source family and physical full-field closure	UNRESOLVED
G9	Policy-gated reveal after qualification	NOT EXECUTED

No gate may upgrade a physical claim merely because Python/C++ numerics agree. An observed negative of a true physical hypothesis is distinguishable from an invalid simulator run.

10 Numerical refinement and error budget

For a fixed nondimensional physical end time $T = 5$, default same-signed reference uses $R = (1, 0.8)$, $z = (-0.15, +0.15)$, $\Gamma = (+1, +1)$, $a/R = 0.1$. Run angular quadratures $n_\varphi \in \{48, 96, 192\}$ at fixed 400 steps, then temporal steps $\{200, 400, 800\}$ with angular resolution held at 96. The preselected reduced-model relative thresholds are 0.02 and 0.01, respectively. A result near a zero crossing requires an absolute scale; no spectral-energy clustering threshold is presently qualified. Spatial *angular-quadrature* refinement is not a 3D Cartesian Euler grid refinement. The core smoothing is a model uncertainty that cannot be reduced away by time refinement.

11 Backend authority and precision

Reference: Python with NumPy float64. Independent C++17 arithmetic kernel: compile separately; executable CLI can be tested on Linux, pybind11 extension is required for certified framework execution on Windows. Framework v1.1.0 requires two of three actual scientific categories (Python, native C++, GPU); the GPU category is absent, and device probes never count. Any precision or source-hash mismatch is blocking. Intel Arc XPU is the default only if a real kernel is added; FP32 and DD32 are screening and not automatically equivalent to binary64.

12 Blindness and reveal

No electron mass, electric charge, Planck constant, experimental spin labels or target-optimized thresholds enter the pilot. The four input configurations are disclosed as procedural controls; they do not constitute an independent blind confirmation study. The official framework's immutable protocol and HMAC-based reveal are only valid *after* the actual generator and freeze; this package does not forge a reveal or use a public-salt key. Private reveal data is excluded from all distributed blind exports.

13 Null distribution and multiplicity

No statistical test of quantized levels is executed. Trajectories are deterministic ODE runs; more timesteps or ring samples do not create independent replicates. The proposed physical-state test requires a preregistered continuum null, truly held-out parameters, a defined spectral test and multiple-testing correction before any such hypothesis is assessed. Current outcomes must not be labelled quantum quantization.

14 Measured results and reproducibility artifacts

Automatic run provenance

Field	Recorded value
Run ID	20261028T225431868035Z-a946cb64
Run phase	BLIND
2-of-3 backend execution	EXECUTION_CONTRACT_PASS
Executed backend categories	py, cpp
GPU provider	intel
Protocol bundle SHA-256 (byte exact)	ea4fe73e55352bd344222bc122cda38380d9f512febb43130de7ea07a33343d6
Portable protocol diagnostic SHA-256	a89ea7f5225dcbbbbc8c8fd64153ed22aac29ee01ff8d8a082318671dcc5b1d5
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	f064d8ea7babdb04365a500bee7066b6ab3c20152264995782fbfc1329a2e834
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Official frozen v1.1.0 runtime integrity?	Framework verified preregistration and embedded runtime bootstrap.
G1	PASS	Analytic source/initial data provenance?	Analytic, deterministic two-ring seed without external dataset.
G2	PASS	Reduced-model numerical and dimensional controls?	Angular-quadrature convergence for reduced ring model only, not 3D Euler.
G3	UNRESOLVED	Target-free spectral discreteness rather than rank swaps?	No blind spectral discrete-level discovery with full-field Euler.
G4	UNRESOLVED	Independent full Euler field/held-out source confirmation?	No independent held-out finite-core field confirmation.
G5	UNRESOLVED	Genuinely executed pybind11 science kernel and parity?	Use framework backend matrix; independently check native pybind11 parity, not a dummy PASS.
G6	NOT_APPLICABLE	Real GPU science kernel with precision authority?	Disabled by framework profile
G7	NOT_APPLICABLE	Cross-source independent replication?	Disabled by framework profile
G8	UNRESOLVED	Finite-core full Euler, physical layer, spinor, morphology gates?	No full 3D Euler, independent toroidal circulation, jet flow, or spinor representation.
G9	DEFERRED	Optional fully qualified blind reveal?	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G3	UNRESOLVED	none
G4	DIAGNOSTIC_ONLY	G3
G5	DIAGNOSTIC_ONLY	G4
G6	NOT_APPLICABLE	G5
G7	NOT_APPLICABLE	G6
G8	DIAGNOSTIC_ONLY	G7
G9	DEFERRED	G8

Numeric backend comparisons (not physical validation)

Backend	Relative L2 error	Result/status
cpp	5.923075954129776e-16	MEASURED_ONLY

Backend manifest


```
{
  "backend_2of3": {
    "artifact": "AUTO_BACKEND_MATRIX.json",
    "counted": 2,
    "executed": [
      "py",
      "cpp"
    ],
    "status": "EXECUTION_CONTRACT_PASS"
  },
  "model": "AXISYMMETRIC_REGULARIZED_FILAMENT_PROXY",
  "note": "P01-P03 local diagnostics; P04-P07 and spinorial claim
unresolved.",
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_authority": "DEVELOPMENT_DIAGNOSTIC_ONLY"
}
```


Source manifest


```
{
  "manifest_sha256":
"bee7a88039d6c7375b5ec2d8eb062c1766a9113ff51c2a15c5290d27e4d8841d",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation, falsifiable next tests and confounders

Classical order swaps, even if numerically convergent, cannot yield electron spin statistics. In particular, changing circulation sign is a classical time-reversal operation at the level of instantaneous induced velocity. The two signs cannot by themselves furnish a nontrivial double-valued rotation representation. Axial transport is a kinematic observable, not an independent spin degree of freedom. A subsequent finite-core Euler study must establish (i) full-field conservation, (ii) core-width and spatial/time convergence separately, (iii) independent toroidal material circulation, (iv) robust nontrivial moduli/levels and (v) a justified state bundle with 2π sign change and 4π return. Sprite/giant-jet analogies are purely morphological; the present incompressible model contains no ionization, electrodynamic radiation or electron ontology.

16 Exact reproducibility commands

```
python -m pytest -q tests
# Linux local independent C++ control:
g++ -O2 -std=c++17 native/ring_cli.cpp -o build/ring_cli
python -m science.run_science --cpp --archive
# Windows local independent C++ / pybind11:
call tools\build_cpp.cmd
# On a separate verified 1.1.0 generator (create-only):
python tools\install_framework_v110.py --framework <FW> ^
  --output <NEW_A###-v0.1.0> --catalog <A###>
# Within generated official instance:
run_all.cmd BACKEND_PREFLIGHT
run_all.cmd ATLAS_VERIFY
python -m pytest -q tests
run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd FULL
run_all.cmd CERTIFY
```

No result from this local pilot represents the above official Framework execution.

17 References

References

[1] H. Helmholtz (1858), "On Integrals of the Hydrodynamical Equations Which Express Vortex Motion," *Journal für die reine und angewandte Mathematik* 55, 25–55. https://doi.org/10.1515/crll.1858.55.25.

[2] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press. https://doi.org/10.1017/CBO9780511624063.

[3] J. Dávila, M. del Pino, M. Musso, J. Wei (2024), "Leapfrogging vortex rings for the three-dimensional incompressible Euler equations," *Communications on Pure and Applied Mathematics* 77, 3843–3957. https://doi.org/10.1002/cpa.22199.

[4] A. J. Niemi (2005), "Exotic Statistics of Leapfrogging Vortex Rings," *Physical Review Letters* 94, 124502. https://doi.org/10.1103/PhysRevLett.94.124502.

[5] D. Finkelstein and J. Rubinstein (1968), "Connection between Spin, Statistics, and Kinks," *Journal of Mathematical Physics* 9, 1762–1779. https://doi.org/10.1063/1.1664510.

[6] T. J. Dekker (1971), "A Floating-Point Technique for Extending the Available Precision," *Numerische Mathematik* 18, 224–242. https://doi.org/10.1007/BF01397083.